

Module 02: Agents — The Robot Job

Learning Objectives

1. Understand that a **Robot Agent** has a Goal (A Job).
2. Learn that robots follow **Rules** (Code).
3. Know the difference between a Tool (Hammer) and an Agent (Robot).

Introduction: The Robot Job

Meet "Clean-Bot 3000". Its goal is: **Clean Floor**. It drives around. If it sees dirt -> It vacuums. If it hits a wall -> It turns around. It doesn't need a human to push it. It is an **Agent**!

A Hammer is just a tool. It does nothing until YOU pick it up. A Robot Agent can do things by itself!

Key Concepts

1. The Goal (The Mission)

Every robot has a mission.

- Mars Rover Mission: Find Rocks.
- Chef Bot Mission: Make Pizza.
- Car Bot Mission: Drive Safe. Without a goal, a robot just sits there.

2. Sensing and Acting

The robot is like a loop.

1. **Sense**: Is there a wall?
2. **Think**: Yes! Stop!
3. **Act**: Turn Left. This loop happens thousands of times a minute!

3. Safety Rules

Robots must be safe. **Rule #1:** Don't hurt humans! **Rule #2:** Don't break yourself! These rules are deep inside the robot's brain.

Activities

Activity 1: Tool vs. Agent

Look at things in your house.

- Broom? (Tool. Needs you).
- Roomba Vacuum? (Agent. Moves itself).
- Pencil? (Tool).
- Cat? (Agent! A biological one).

Activity 2: The "If-Then" Game

Play with a friend. Give them a rule:

- "IF I clap, THEN you jump."
- "IF I stomp, THEN you sit." Test their programming!

Summary

An Agent is a machine that has a goal and can act to reach it. Robots are special agents built by humans to help us do work!

References

- *R is for Robot: A Noisy Alphabet* by Adam F. Watkins
- *Wall-E* (Movie about a robot with a job)